Metabolic Assessment Form™ – Blood Chemistry Key

CATEGORY I: Colon

Eosinophils **increased**

Check immune health
Consider ordering a GI stool test or Cyrex™ Laboratories Array 2 for intestinal permeability

CATEGORY II: Intestinal Barrier

Protein and globulin **increased or decreased**

CATEGORY III: Chemical Tolerance

Protein and globulin **increased or decreased** associated with intestinal permeability
Depressed 25-hydroxyvitamin D levels associated with reduced regulatory T-cell support

CATEGORY IV: Stomach

(Check for Hypochlorhydria)

Protein and globulin **increased or decreased**
BUN **increased or decreased**
Phosphorus **decreased**

Check for possible H. pylori infection, which may alter acid production:

WBC **increased or decreased**
Neutrophils **increased**

CATEGORY V: Stomach

(Check for Hyperacidity)

Protein and globulin **increased or decreased**
BUN **increased or decreased**
Phosphorus **decreased**

Check for possible H. pylori infection, which may alter acid production:

WBC **increased or decreased**
Neutrophils **increased**

CATEGORY VI: Pancreas

Pancreatic enzyme deficiency shows up many times as insulin resistance

Check for pancreatic enzyme deficiency
Check Category XI on Metabolic Assessment Form™ results to assess the peripheral utilization of sugars

Usually only symptoms associated and not associated with blood chemistry markers

CATEGORY VII: Small Intestine

Usually only symptoms associated and not associated with blood chemistry markers

CATEGORY VIII: Biliary

GGTP, alkaline phosphatase—don't usually show up until obstruction

MAF symptoms may be due to thick bile, pre-gallstones, or gallstones

Some individuals may have no symptoms because they avoid foods that cause them

CATEGORY IX: Hepatic Detoxication

Usually only symptoms associated and not associated with blood chemistry markers

May be suggested from functional elevation of SGOT, SGPT, and/or GGTP

CATEGORY X: Sugar Metabolism

(Check for Hypoglycemia)

Glucose **decreased**
LDH **decreased**

Symptom groups listed on this form are not intended to be used for a diagnosis of any disease or condition.

The lab tests and blood markers referenced on this form are solely intended for nutritional evaluation within the reference range. Blood test markers that are out of range may require pharmaceutical intervention or other medical treatment and will require that the individual be seen by a medical practitioner with a scope of practice to diagnose, treat, and cure disease.

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CATEGORY XI: Peripheral Utilization of Sugars

(Check for Insulin Resistance)

Glucose **increased**
Triglycerides **increased**
Total cholesterol and LDL **increased**
HDL **decreased**
Uric acid **increased**

CATEGORY XII: Adrenal

(Check for Hypofunction)

Sodium and potassium **changes**

Check Category X on Metabolic Assessment Form™ results to assess sugar metabolism

CATEGORY XIII: Adrenal

(Check for Hyperfunction)

Sodium and potassium **changes**

Check Category XI on Metabolic Assessment Form™ results to assess the peripheral utilization of sugars

CATEGORY XIV: Electrolyte and pH Balance

Sodium, potassium, chloride, calcium, and magnesium levels
CO_2 levels
Anion gap

CATEGORY XV: Thyroid

(also check for Hypofunction)

TSH **increased**
T3 Uptake
Total T4
Total T3
FTI
Thyroid antibodies **increased**

CATEGORY XVI: Thyroid

(Check for Hyperfunction)

TSH **decreased**
T3 Uptake
Total T4
Total T3
FTI
Thyroid antibodies **increased**

CATEGORY XVII: Prostate (Males Only)

PSA
Free PSA

CATEGORY XVIII: Andropause (Males Only)

Check for insulin resistance
Check for systemic inflammation
Check for GI infections
Check for neurodegeneration
Check hormone levels (testosterone:estrogen ratio)

CATEGORY XIX: Menstruating Women

Check for systemic inflammation
Check hormone levels

CATEGORY XX: Menopausal Women

Check for systemic inflammation
Check hormone levels

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Metabolic Assessment Form™ Interpretation

The Metabolic Assessment Form™ (MAF) is a tool you can use as part of your intake and assessment process. It is organized to help you quickly and easily identify issues and their subjective importance to your patients. The MAF is used to develop dietary, nutritional, and lifestyle strategies, helps determine which blood tests should be ordered, correlates the patient's symptoms to blood chemistry laboratory patterns, and helps monitor the patient's outcomes. It is not designed to offer a diagnosis for any condition. Further determinations will be necessary to evaluate the value of the findings from this form.

The MAF Keys

The MAF has two keys, the MAF Blood Chemistry Key and the MAF Nutritional Key. The blood chemistry key notes blood test markers related to the category and may offer suggestions for additional testing.

The nutritional key suggests protocols for each category. The nutritional protocols are used to support various body systems. **The products listed are only intended as support for natural metabolic processes that are under temporary stress or to address an additional demand for nutrients. These recommendations have not been evaluated by the Food and Drug Administration and should not be construed as claims to treat, cure, or prevent disease, as they are not intended to act as drugs or to replace any drug prescribed by a physician.** We believe that such an act may be detrimental to the health and well-being of an individual and that dietary supplements should only be used in a supportive role in such conditions.

The MAF Parts

The assessment form has four parts that the patient completes, which includes:

- Part I: Health Concerns – The patient lists five major health concerns
- Part II: Categories – Gather subjective data to prioritize and recognize system concerns for different metabolic categories
- Part III: Foods – Recognize dietary habits that may alter metabolism
- Part IV: Medications – Recognize medications that may alter metabolism

Part I: Health Concerns

Part I is the most important category of the questionnaire form, as it discusses an individual's five major health concerns in order of importance. This helps the clinician to understand what the individual expects from the clinician. The clinician can also use this section to help the individual understand what systems are associated with his or her symptoms.

Part II: Categories

The questions in each category relate to a body part, function, or physiological mechanism. The symptoms are subjective. The categories are:

- Colon
- Intestinal Barrier
- Chemical Tolerance
- Stomach (Two Categories)
- Pancreas
- Small Intestine
- Biliary
- Hepatic Detoxication
- Sugar Metabolism
- Peripheral Utilization of Sugars
- Adrenal (Two Categories)
- Electrolyte and pH Balance
- Thyroid (Two Categories)
- Prostate
- Andropause (Males Only)
- Menstruating Women
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